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Harmonic Constituents for 9447130, Seattle WA

Constituent

Order in which the National Ocean Service lists the constituents

Name

Common name used to refer to a particular constituent, subscript refers to the number of cycles per day

Amplitude

One-half the range of a tidal constituent

Phase

The phase lag of the observed tidal constituent relative to the theoretical equilibrium tide

Speed

The rate change in the phase of a constituent, expressed in degrees per hour. The speed is equal to 360 degrees divided by the constituent period expressed in hours

Showing Harmonic Constituents for

9447130 Seattle

Data Units ☒ Feet
☐ Meters

Timezone ☒ Local
☐ GMT

Description

The full name of the tidal constituent

Please refer to the [Tide and Current Glossary \(/glossary.html\)](/glossary.html) for definitions of terms. Amplitudes are in feet. Phases are in degrees, referenced to local. Z_0 (MSL): 0 feet

Constituent					
#	Name	Amplitude	Phase	Speed	Description
1	M2	3.487	138.9	28.984104	Principal lunar semidiurnal constituent
2	S2	0.879	156.8	30.0	Principal solar semidiurnal constituent
3	N2	0.702	113.6	28.43973	Larger lunar elliptic semidiurnal constituent
4	K1	2.736	156.5	15.041069	Lunar diurnal constituent
5	M4	0.069	97.0	57.96821	Shallow water overtides of principal lunar constituent
6	O1	1.506	143.1	13.943035	Lunar diurnal constituent
7	M6	0.03	337.2	86.95232	Shallow water overtides of principal lunar constituent
8	MK3	0.118	87.1	44.025173	Shallow water terdiurnal
9	S4	0.007	134.3	60.0	Shallow water overtides of principal solar constituent
10	MN4	0.03	73.3	57.423832	Shallow water quarter diurnal constituent
11	NU2	0.144	127.4	28.512583	Larger lunar evectional constituent
12	S6	0.0	0.0	90.0	Shallow water overtides of principal solar constituent
13	MU2	0.112	15.1	27.968208	Variational constituent
14	2N2	0.075	89.9	27.895355	Lunar elliptical semidiurnal second-order constituent
15	OO1	0.102	201.1	16.139101	Lunar diurnal
16	LAM2	0.066	174.3	29.455626	Smaller lunar evectional constituent
17	S1	0.069	285.0	15.0	Solar diurnal constituent

18	M1	0.079	188.2	14.496694	Smaller lunar elliptic diurnal constituent
19	J1	0.141	188.7	15.5854435	Smaller lunar elliptic diurnal constituent
20	MM	0.0	0.0	0.5443747	Lunar monthly constituent
21	SSA	0.079	216.3	0.0821373	Solar semiannual constituent
22	SA	0.23	282.9	0.0410686	Solar annual constituent
23	MSF	0.0	0.0	1.0158958	Lunisolar synodic fortnightly constituent
24	MF	0.049	148.2	1.0980331	Lunisolar fortnightly constituent
25	RHO	0.049	137.2	13.471515	Larger lunar evectional diurnal constituent
26	Q1	0.239	141.7	13.398661	Larger lunar elliptic diurnal constituent
27	T2	0.052	158.4	29.958933	Larger solar elliptic constituent
28	R2	0.01	130.8	30.041067	Smaller solar elliptic constituent
29	2Q1	0.033	162.7	12.854286	Larger elliptic diurnal
30	P1	0.843	156.5	14.958931	Solar diurnal constituent
31	2SM2	0.026	36.3	31.015896	Shallow water semidiurnal constituent
32	M3	0.013	190.2	43.47616	Lunar terdiurnal constituent
33	L2	0.161	182.5	29.528479	Smaller lunar elliptic semidiurnal constituent
34	2MK3	0.115	65.1	42.92714	Shallow water terdiurnal constituent
35	K2	0.259	157.0	30.082138	Lunisolar semidiurnal constituent
36	M8	0.003	356.9	115.93642	Shallow water eighth diurnal constituent
37	MS4	0.039	117.4	58.984104	Shallow water quarter diurnal constituent

Tidal Harmonic Constants Source: Vector Averaged from 5 one year analyses 2000-2024. SSA and SA from 20 year analyses (2000-2024).

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